

PAPER_695: NGC 7635 Bubble Nebula: O-Star Wind Driven Cavity and UQFF

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Abstract

NGC 7635 (Bubble Nebula) is a circumstellar nebula driven by the O-star BD+60°2522. The Weaver et al. (1977) analytic model for wind-blown cavities is applied with UQFF modifications to the terminal wind velocity and bubble expansion.

1. Parameters

Parameter	Value
M_*	$44M_\odot$
L_*	$4 \times 10^5 L_\odot$
$\dot{M}_{wind}$	$10^{-5} M_\odot/\text{yr}$
v_∞	2000 km/s
R_{bubble}	5 ly

2. Wind Luminosity

$$L_w = \frac{1}{2} \dot{M} v_\infty^2$$

3. Bubble Expansion (Weaver 1977)

$$R_b(t) = 0.88 \left(\frac{L_w}{\rho_0} \right)^{1/5} t^{3/5}$$

4. UQFF Wind Velocity

$$v_{UQFF} = v_\infty (1 + f_{TRZ}) \sqrt{\frac{\rho_{UA}}{\rho_{SCm}}}$$

Enhanced by factor $\sqrt{\rho_{UA}/\rho_{SCm}} = \sqrt{10} \approx 3.16$.

5. Shock Compression (Strong Shock)

$$\frac{\rho_2}{\rho_1} = \frac{\gamma + 1}{\gamma - 1} = 4 \quad (\gamma = 5/3)$$

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$$\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$$

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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 $f_{U_b} \cdot$
 $(1 - e^{-[SSq] \cdot m/M_{\odot}}).$
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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Weaver et al. (1977), ApJ, 218, 377
- UQFF Framework, Star-Magic Session 174

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